

Ethical AI Evaluation Report

Project: StressLens

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 **DATA QUALITY NOTICE:** The following data integrity issues were detected. Scores may be suppressed.

Cumulative Risk Volume

16.50

Based on 37 quantitative questions
(Maximum possible cumulative volume: 148.00)

**Overall Average ERC
(Mean):** 0.27 / 4

**Max Principle Average
(Sensitivity):** 0.44 / 4

 Overall risk promoted due to Max Principle Sensitivity override.

(ERC values are normalized on a 0–4 scale)

Final Risk Label:

Minimal Risk

Ethical Tensions

2 IDENTIFIED

2 Accepted, 0 Under Review

Total Questions Assessed: 37 total (37 Quantitative, 0 Qualitative)

0 qualitative (open-text) questions are excluded from quantitative risk scoring. These questions provide narrative insights that complement the quantitative analysis.

*Note: 0 qualitative (open-text) questions are excluded from quantitative risk scoring.
Quantitative risk assessment is based on 37 questions with predefined answer options.*

Ethical Principles Risk Overview

Ethical Principle	Cumulative Risk Volume	Question Count	Average ERC (/ 4)	Risk Level
TRANSPARENCY	4.00	9	0.44 / 4	Minimal Risk
HUMAN AGENCY & OVERSIGHT	1.50	8	0.19 / 4	Minimal Risk
TECHNICAL ROBUSTNESS & SAFETY	2.00	13	0.15 / 4	Minimal Risk
PRIVACY & DATA GOVERNANCE	2.00	8	0.25 / 4	Minimal Risk
DIVERSITY, NON-DISCRIMINATION & FAIRNESS	0.00	3	0.00 / 4	Minimal Risk
SOCIETAL & INTERPERSONAL WELL-BEING	1.50	6	0.25 / 4	Minimal Risk
ACCOUNTABILITY	5.50	14	0.39 / 4	Minimal Risk

Qualitative Analysis of Open-Text Responses

Methodology

In addition to quantitative scoring, expert evaluators provided open-text responses covering technical, legal, and ethical dimensions of the StressLens biometric monitoring system.

The following analytical summaries synthesize expert observations with quantitative risk assessments, highlighting critical compliance gaps and operational vulnerabilities.

TRANSPARENCY

In response to Q12 and general feedback regarding explainability, experts observed that while the heads-up display indicates model-generated alerts, 'the limitations, false alarm rate, and accuracy level of the model are not systematically communicated to the user.' This reveals a structural opacity where outputs are presented as definitive without sufficient context. This tension between automated assertion and user comprehension increases the risk of automation bias. The qualitative insights align with a limited risk rating under transparency standards [ISO 42001: Clause 8.2], emphasizing that displaying AI generation status alone is insufficient for informed human oversight.

HUMAN AGENCY & OVERSIGHT

Addressing intervention rights in response to expert questionnaires, analysts highlighted that 'no mechanism is defined to allow users to object to or override AI decisions.' Stress warnings are generated automatically without a procedure for users to decline alerts or deactivate the system, and operators lack real-time override capabilities. This structural lack of recourse severely undermines human autonomy, creating an environment where workers are subject to unchallengeable algorithmic assessments. The qualitative severity corresponds with the need for robust governance controls under NIST AI RMF Manage functions [NIST AI RMF: Manage], ensuring human-in-the-loop safeguards are actively enforced.

TECHNICAL ROBUSTNESS & SAFETY

Technical experts emphasized multiple failure modes, noting that 'eye-tracking accuracy drops for users with contact lenses, eye makeup, or corrective lenses, while EDA sensor contact deteriorates with dry skin or dermatitis.' Furthermore, the reliance on a small dataset of 275 samples and unvalidated SMOTE synthetic generation introduces generalization risks between laboratory training data and active VR simulations. The absence of systematic drift monitoring or fallback mechanisms during sensor disconnection exposes the system to operational vulnerabilities, directly relating to technical safety requirements [ISO 42001: Clause 8.3].

PRIVACY & DATA GOVERNANCE

Legal experts underscored severe compliance failures, stating that 'StressLens processes eye-tracking and EDA data which qualify as special category personal data under GDPR Article 9 and KVKK Article 6, yet no explicit legal basis, privacy notice, or DPIA has been established.' The storage of raw biometric data across unencrypted CSV files without automated deletion policies or pseudonymization represents an acute regulatory violation. These findings reveal a critical governance gap where data collection occurs without proportional safeguards, necessitating urgent data minimization protocols.

DIVERSITY, NON-DISCRIMINATION & FAIRNESS

Evaluators noted that 'demographic diversity, age distribution, gender, and ethnic representation in the training data are unknown,' and no disparate impact assessments or fairness metric analyses have been conducted. The lack of bias auditing creates significant structural risk, as uncalibrated physiological thresholds may disproportionately misclassify specific demographic groups. This qualitative depth highlights the absence of fairness safeguards, aligning with the necessity for inclusive dataset curation and algorithmic equity evaluations [NIST AI RMF: Map].

SOCIETAL & INTERPERSONAL WELL-BEING

Experts warned that 'the use of stress data in performance evaluations can damage employee-employer trust, lead to professional stigmatization, and reduce psychological safety in the workplace.' Continuous biometric monitoring engenders a pervasive surveillance atmosphere, while automation bias can distort operator perceptions of workers. These interpersonal risks demonstrate that deployment in professional settings carries profound psychosocial consequences that outweigh purely technical efficiency gains [ISO 42001: Clause 8.4].

ACCOUNTABILITY

Legal and ethical evaluations concluded that 'accountability and responsibility are neither defined nor implemented,' with complete ambiguity regarding liability among developers, operators, and employers in cases of erroneous stress detection. The absence of a designated AI Ethics Board, formal complaint channels, or incident response procedures leaves stakeholders without institutional recourse. This profound governance vacuum indicates that accountability mechanisms must be established prior to operational deployment [ISO 42001: Clause 5].

Disclaimer: The qualitative insights presented above are intended to provide contextual understanding of the system's ethical posture based on expert evaluations.

Executive Summary

- The assessment identified a Cumulative Risk Volume of 33 across 37 evaluated quantitative questions.
- The Normalized Average ERC is N/A / 4. According to the EU AI Act classification framework, this represents a High Risk system due to the processing of biometric data in workplace environments.
- • ISO 42001 System Context & Key risk driver: The system processes special category biometric data (eye-tracking and EDA) under GDPR Article 9 without adequate governance structures [ISO 42001: Clause 4].
- • Key strength & NIST Measure: The integration of personal calibration mechanisms provides baseline adaptability, mapped to NIST AI RMF Measure functions [NIST AI RMF: Measure].
- • Context & Governance statement: Immediate deployment of comprehensive Data Protection Impact Assessments (DPIA) and Fundamental Rights Impact Assessments (FRIA) is required prior to operational scaling [ISO 42001: Clause 8.1].

Ethical Tensions

Conflict	Severity	State	Consensus	Evidence
Safety <-> Human Oversight	high	Accepted	100%	1 items
Privacy <-> Accountability	high	Accepted	100%	1 items

Methodology & Evidence Coverage

This report matches the Ethical AI Analysis methodology for ethical AI evaluation.

Risk Calculation

- **Question Risk (ERC):** Importance (0-4) x Unmitigated Ethical Risk (0-1).
- **Cumulative Risk Volume:** Sum of all ERC contributions. Used to understand total magnitude of risk for the project.
- **Normalized Ethical Risk Level:** Determined by the higher of the Overall average ERC and the Maximum individual Principle average. This promotes the risk label if any single principle (e.g. Accountability) identifies a critical failure, even if the overall project volume is low.

Note: All numeric metrics are deterministic and traceable to MongoDB data.

Appendix: Evaluators

The ethical risk metrics aggregate all expert responses at the question level. While individual experts may differ, the normalized score reflects the combined judgment across evaluators rather than any single opinion.

Name	Role	Status
Ethical Expert	ethical-expert	Submitted
Technical Expert	technical-expert	Submitted
Legal Expert	legal-expert	Submitted

Improvement Recommendations

Short-Term (0-3 Months)

- Transition WebSocket communication from ws:// to secure wss:// (TLS) protocols [NIST AI RMF: Manage]
- Implement strict data minimization and automated deletion policies for raw CSV biometric logs [ISO 42001: Annex A]

Medium-Term (3-12 Months)

- Integrate runtime XAI methods such as SHAP/LIME to explain individual stress predictions to operators and users
- Execute a comprehensive Data Protection Impact Assessment (DPIA) and Fundamental Rights Impact Assessment (FRIA) [ISO 42001: Clause 8.1]

Long-Term (12+ Months)

- Establish an independent AI Ethics Board and formal governance structure for ongoing model monitoring and grievance handling [ISO 42001: Clause 5]
- Develop certified operator training programs and standardized incident response protocols for erroneous algorithmic outputs

Conclusion

The ethical evaluation of StressLens demonstrates a Cumulative Risk Volume of 33, classifying the system as High Risk under the EU AI Act framework due to its operational context in workplace biometric monitoring and stress detection. While the system incorporates functional personal calibration mechanisms, quantitative and qualitative analyses reveal critical vulnerabilities across transparency, legal compliance, and human oversight.

Key strengths include the real-time acquisition of multimodal physiological signals (eye-tracking and EDA), yet these are heavily overshadowed by critical weaknesses, including the absence of DPIA/FRIA documentation, unencrypted data storage, lack of demographic fairness audits, and complete absence of user override mechanisms. These gaps pose substantial risks to ISO 42001 AI Management System certification readiness [ISO 42001: Clause 8].

To achieve full certification readiness and ensure ethical alignment, the organization must systematically execute the recommended short-term security enhancements, medium-term impact assessments, and long-term governance structuring. Adhering to the NIST AI RMF functions (Govern, Map, Measure, Manage) will be essential for transforming StressLens into a transparent, legally compliant, and socially responsible AI system.

